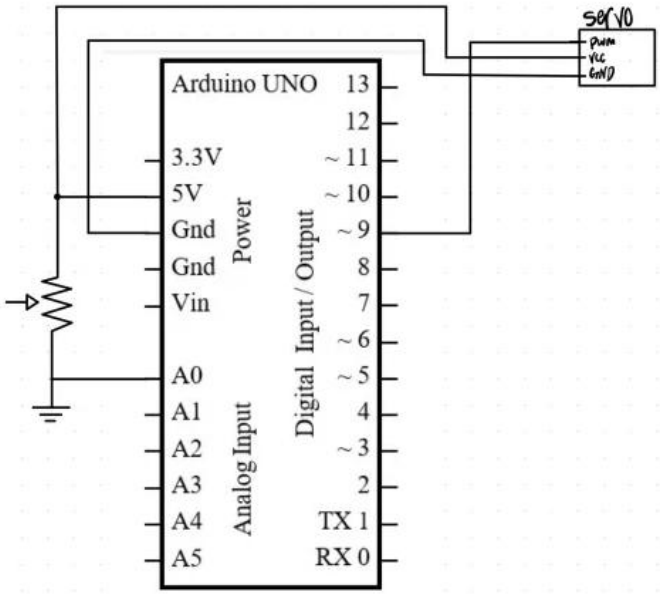
ECE 2305 – Introduction to C Programming
Hardware Project 05

Select one of the following projects and document your work with the following sections. Provide a brief description of the system and how you are designing it to operate.

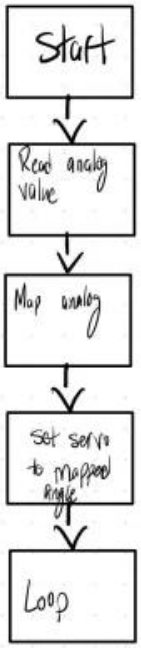
A. Hardware Diagram:

Provide a hardware diagram of the components.



B. Program Flowchart:

Draw a flowchart of the program.



C. Arduino Source Code

Insert the Arduino Source Code into the document.

```
// ECE 2305 - Hardware Project 05

#include <Servo.h>

Servo voltmeterServo;
const int potPin = A0; // Potentiometer connected to A0

// Function to smooth noisy analog readings
int readStableAnalog(int pin) {
  long sum = 0;
  for (int i = 0; i < 10; i++) {
    sum += analogRead(pin);
    delay(1);
  }
  return sum / 10;
}

void setup() {
  voltmeterServo.attach(9); // Servo connected to D9
}

void loop() {
  int sensorValue = readStableAnalog(potPin);         // Read stable analog value
  int angle = map(sensorValue, 0, 1023, 0, 180);       // Map 0–1023 to 0–180

  static int lastAngle = -1;
  if (abs(angle - lastAngle) > 5) {                   // Only update if angle
changes > 2°
    voltmeterServo.write(angle);                      // Move servo
    lastAngle = angle;
  }

  delay(20);  // Smooth motion, reduce jitter
}
```

D. Demonstration Video

Record and upload a video demonstrating the operation of the circuit. **DONE**

Save the document as a *PDF* file and submit the *PDF* document to *Blackboard*.

ECE 2305 – Introduction to C Programming
Hardware Project 05

1. Servo Voltmeter

Program an Arduino board to accept an analog input voltage and display the magnitude of the voltage using a Hobby Servo to display the voltage. Use the servo horn to direct an arrow to the appropriate voltage. Provide as much precision in the display as possible.

